THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 40	Area $= \frac{1}{2} \times$ base $\times$ height.
2	**(c)** $\frac{5}{6}$	Convert to a common denominator first.
3	**(d)** do not touch (separate)	Compare 9 with 5+3=8 and |5-3|=2.
4	**(b)** 2	Along its two diagonals.
5	**(a)** -5	Same signs: add and keep sign; different signs: subtract.
6	**(d)** 24	Half the product of the legs.
7	**(b)** $\frac{5}{12}$	Common denominator first.
8	**(a)** 4	Midpoint of AB.
9	**(c)** 72	$360° \div$ number of sides.
10	**(a)** -7	Add the opposite.
11	**(d)** 32	Find the side first.
12	**(b)** $\frac{5}{8}$	Cross-multiply or use a common denominator.
13	**(b)** 5	Diagonals of a rectangle are equal and bisect each other.
14	**(d)** the same letter	Vertical-axis symmetry keeps it the same.
15	**(c)** -6	Right is positive, left is negative.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Area $= \frac{1}{2} \times b \times h$ (1); $\frac{1}{2} \times 12 \times 9 = 54$ cm^2 (1).	1+1
17	LCM of 4 and 6 is 12 (1); $\frac{5}{12}$ ($\frac{5}{12}$) (1).	1+1
18	Sum of radii $= 10$, difference $= 2$ (1); The circles touch externally (1).	1+1
19	Fold the figure so that the two halves coincide; There are 2 lines of symmetry (1); Lines pass through vertices/midpoints as per the shape (1).	1+1
20	5 yellow tokens and 3 red tokens (1); Cancel pairs: the value is 2 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Perimeter $= 2(15 + 8) = 46$ cm (1); Area $= 120$ cm^2 (1); Units must be written (1).	1+1+1
22	LCM of denominators $= 12$ (1); Sum $= \frac{17}{12}$ (1); Greater than 1 (1).	1+1+1
23	Draw AB = 6 cm (1); With A and B as centres and the same radius more than half of AB, draw arcs on both sides (1); Join the intersection points; the line meets AB at its midpoint, which is at 3.0 cm from A (1).	1+1+1
24	Sides $= 360 \div 40 = 9$ (1); Lines of symmetry $= 9$ (1); Order of rotational symmetry equals the number of sides (1).	1+1+1
	OR: First two: -4 (1); Then add 5 (1); Result 1 (1).	1+1+1
25	Legs 6 cm and 8 cm; area $= \frac{1}{2} \times 6 \times 8 = 24$ cm^2 (1); Perimeter $= 6 + 8 + 10 = 24$ cm (1); Units: cm and cm^2 (1).	1+1+1
	OR: Pieces $= 3 \div$ \frac{2}{5} (count $\frac{15}{2}$) (1); Whole pieces 7 (1); Left over $\frac{1}{5}$ m (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Area of first part $= 10 \times 6 = 60$ cm^2 (1); Area of second part $= 4 \times 5 = 20$ cm^2 (1); Total area $= 80$ cm^2 (1); Area after removing smaller part $= 60$ cm^2 (1); State the units (1).	1+1+1+1+1
	OR: Total =\frac{1}{4}+\frac{1}{3}= $7/12$ (1); Left $= 1 - 7/12 = 5/12$ (1); More on Tuesday (1); Check total plus left equals 1 (2).	1+1+1+2

27	Draw AB = 8 cm (1); Draw perpendiculars at A and B and mark 6 cm on each (1); Join the free ends (1); Draw both diagonals and measure: AC = BD (1); Diagonals bisect each other so OA = OC = AC/2 (1).	1+1+1+1+1
	OR: (a) 6 (1); (b) 6 (1); (c) 60° (1); (d) $4 \times 60 = 240^\circ$ (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** 9	Breadth = area ÷ length.
2	**(a)** $3\frac{5}{6}$	Add the whole parts and fraction parts separately.
3	**(c)** 60	Triangle OAB is isosceles with OA = OB.
4	**(a)** 120	Smallest turn.
5	**(c)** 8	Count the integers strictly between.
6	**(d)** 1200	Cost = perimeter × rate.
7	**(c)** $\frac{2}{3}, \frac{3}{4}, \frac{5}{6}$	Use a common denominator.
8	**(c)** 4	Subtract the radii.
9	**(c)** 7	Sides = lines of symmetry.
10	**(c)** -9	Add the rise.
11	**(b)** 300	Divide the floor area by the area of one tile.
12	**(b)** 3	Reduce the fraction.
13	**(d)** equal and perpendicular bisectors of each other	Property
14	**(c)** Z	Z fits after a half turn but cannot be folded.
15	**(d)** 0	$n + (-n) = 0$.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Rectangle area = 60 cm^2 (1); Each triangle = 30 cm^2 (1).	1+1
17	LCM of 6 and 3 is 6 (1); $\frac{1}{2}$ (1).	1+1
18	Diagonals are equal and bisect each other (1); OA = OB = 6 cm; BD = 12 cm (1).	1+1
19	Angle = $360 \div 5$ (1); 72° and order 5 (1).	1+1
20	-4 is to the right of -7 on the number line (1); $-4 > -7$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Diagonals of a square are perpendicular (1); $\angle AOB = 90^\circ$ (1); Triangle AOB is isosceles right-angled so $\angle OAB = 45^\circ$ (1).	1+1+1
22	(i) An isosceles triangle or the letter A (1); (ii) The letter Z or S (1); Example check: A has a vertical line (1).	1+1+1
23	At noon: $-4 + 9 = 5$°C (1); In the evening: $5 - 8 = -3$°C (1); Below 0°C (1).	1+1+1
24	Floor = 400 cm by 300 cm (1); Tiles along length 20, along width 15 (1); Number of tiles = 300 (1).	1+1+1
	OR: Multiply by 4 to get ? = 12 (1); For the second, denominator = 15 (1); Both are equal to $\frac{3}{5}$ (1).	1+1+1
25	Diameter = 8 cm (1); Longest chord = 8 cm (1); No: 9 cm exceeds the diameter (1).	1+1+1
	OR: Image is the same distance behind the mirror (1); Image is 3 units below the line (1); Distance = 6 units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	After rising: $-80 + 30 = -50$ m (1); After diving: $-50 - 25 = -75$ m (1); Total distance travelled = $30 + 25 = 55$ m (1); Below sea level (2).	1+1+1+2
	OR: Outer rectangle: 24 m by 19 m (1); Outer area = 456 m^2 (1); Garden area = 300 m^2 (1); Path area = 156 m^2 (1); State units (1).	1+1+1+1+1
27	After pouring: $3\frac{5}{6}$ litres (1); Subtract $\frac{1}{4}$ (1); Left $3\frac{7}{12}$ litres (1); Check with improper fractions (2).	1+1+1+2

	OR: AB = 7 (1); Sum of radii $= 9$; difference $= 1$ (1); They intersect at two points (1); Common points: 2 (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 300	1 m = 100 cm.
2	**(b)** $\frac{3}{4}$	Equal sharing: divide 3 by the number of children.
3	**(b)** 2	Two pairs of equal opposite sides.
4	**(c)** 8	Order $= 360° \div$ angle.
5	**(a)** 50	Debt is negative.
6	**(d)** 42	Area $= \frac{1}{2} \times$ base $\times$ height.
7	**(c)** $\frac{11}{12}$	Convert to a common denominator first.
8	**(a)** 30	$2(l + b)$.
9	**(a)** 0	Symmetry property
10	**(d)** -17	Same signs: add and keep sign; different signs: subtract.
11	**(b)** 30	Half the product of the legs.
12	**(a)** $\frac{1}{3}$	Common denominator first.
13	**(b)** touch externally	Compare 8 with 5+3=8 and |5-3|=2.
14	**(a)** 60	$360° \div$ number of sides.
15	**(a)** 5	Add the opposite.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Perimeter $= 2(30 + 20) = 100$ m (1); Cost $= 100 \times 15 =$ ₹1500 (1).	1+1
17	$2\times$5+3=13 (1); $13/5$ (1).	1+1
18	Draw the perpendicular bisector with compass arcs of the same radius above 'half AB' (1); The point is 3 cm from A (1).	1+1
19	Reverse the order of letters and reflect each letter; Reversed order: MAM (1); Looks the same (1).	1+1
20	$7 - 3 = 4$ (1); $3 - 7 = -4$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$5 - 9 = -4$ (1); $-4 + (-3) = -7$ (1); Check using the number line (1).	1+1+1
22	Perimeter $= 2(21 + 10) = 62$ cm (1); Area $= 210$ cm^2 (1); Units must be written (1).	1+1+1
23	LCM of denominators $= 10$ (1); Sum $= \frac{9}{10}$ (1); Less than 1 (1).	1+1+1
24	Draw AB = 7 cm (1); With A and B as centres and the same radius more than half of AB, draw arcs on both sides (1); Join the intersection points; the line meets AB at its midpoint, which is at 3.5 cm from A (1).	1+1+1
	OR: Sides $= 360 \div 45 = 8$ (1); Lines of symmetry $= 8$ (1); Order of rotational symmetry equals the number of sides (1).	1+1+1
25	First two: -10 (1); Then add 15 (1); Result 5 (1).	1+1+1
	OR: Legs 5 cm and 12 cm; area $= \frac{1}{2} \times 5 \times 12 = 30$ cm^2 (1); Perimeter $= 5 + 12 + 13 = 30$ cm (1); Units: cm and cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Letters with vertical symmetry: M, A, T, H (1); Letters with horizontal symmetry: H (1); Order-2 rotational letters: H (1); Word symmetric (vertical): no (2).	1+1+1+2

	OR: Total $= 120 + (-80) + 50 + (-30) = 60$ (1); Best day: Mon (₹120) (1); Worst day: Tue (₹-80) (1); Positive total means overall profit (2).	1+1+1+2
27	Area of first part $= 12 \times 5 = 60$ cm^2 (1); Area of second part $= 6 \times 4 = 24$ cm^2 (1); Total area $= 84$ cm^2 (1); Area after removing smaller part $= 60$ cm^2 (1); State the units (1).	1+1+1+1+1
	OR: Total $=$\frac{2}{5}+\frac{1}{4}$= 13/20$ (1); Left $= 1 - 13/20 = 7/20$ (1); More on Monday (1); Check total plus left equals 1 (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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